

# CSV Statistics Analyzer

A C++17 command-line data science tool

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## Project summary

This project implements low-level CSV loading, validation, cleaning, descriptive statistics, aligned console reporting, and CSV export without relying on a statistics library. It accepts the Iris dataset or another header-based CSV containing numeric columns.

## Implementation decisions

Quoted fields and escaped quotes are supported. Empty, malformed, and non-finite numeric cells are counted and skipped. Sample standard deviation uses  $n - 1$ . Quartiles use linear interpolation at  $(n - 1)p$ . Fully categorical columns are excluded.

# Iris summary statistics

| Feature      | Count | Invalid | Min | Max | Mean  | Median | Mode | Sample SD | Q1  | Q3  |
|--------------|-------|---------|-----|-----|-------|--------|------|-----------|-----|-----|
| Sepal length | 150   | 0       | 4.3 | 7.9 | 5.843 | 5.8    | 5.0  | 0.828     | 5.1 | 6.4 |
| Sepal width  | 150   | 0       | 2.0 | 4.4 | 3.057 | 3.0    | 3.0  | 0.436     | 2.8 | 3.3 |
| Petal length | 150   | 0       | 1.0 | 6.9 | 3.758 | 4.35   | 1.4  | 1.765     | 1.6 | 5.1 |
| Petal width  | 150   | 0       | 0.1 | 2.5 | 1.199 | 1.3    | 0.2  | 0.762     | 0.3 | 1.8 |

## Independent verification

A pandas script reproduced all C++ results to six decimal places. pandas std() and Excel STDEV.S match the analyzer because all use sample standard deviation; STDEV.P would be slightly smaller.

## Robustness test

A five-row test CSV contained an empty value, non-numeric text, and a quoted field containing a comma. The analyzer completed without crashing, reported 2 invalid cells in the first numeric column and 1 in the second, and excluded the categorical label column.

# Interpretation

The analyzer processed all 150 observations in the Iris dataset and found four numeric columns: sepal length, sepal width, petal length, and petal width. No missing or malformed values appeared in these columns, so every statistic uses the full dataset. The categorical species column was correctly excluded from numeric calculations.

Petal length varies the most in absolute terms. Its sample standard deviation is approximately 1.765, compared with 0.828 for sepal length, 0.436 for sepal width, and 0.762 for petal width. Its range is also the widest: 1.0 to 6.9, a spread of 5.9 units. This large variation is expected because the dataset combines three Iris species whose petal measurements differ substantially. Petal width has a smaller absolute standard deviation but a broad range relative to its mean, from 0.1 to 2.5.

Mean and median provide a simple indication of distribution shape. Sepal length has a mean of 5.843 and a median of 5.8, while petal length has a mean of 3.758 and a median of 4.35. The noticeably lower mean for petal length suggests that its distribution is pulled toward smaller values. However, this should not be interpreted as ordinary skewness alone: the combined dataset is multimodal because each species forms a different measurement cluster. Petal width shows a similar pattern, with a mean of 1.199 and median of 1.3. Sepal width has a mean of 3.057 and median of 3.0, indicating comparatively little separation between these two center measures.

The interquartile ranges reinforce these observations. Petal length spans 1.6 to 5.1 in the middle 50% of records, much wider than sepal width's 2.8 to 3.3. The modes are 5.0, 3.0, 1.4, and 0.2 respectively. Because several values may tie for highest frequency, this program reports the smallest modal value, matching the first mode returned by pandas.

Verification with pandas produced the same results to six decimal places. The program uses sample standard deviation, dividing by  $n - 1$ , which matches pandas `std()` and Excel `STDEV.S`. Excel `STDEV.P` would instead calculate population standard deviation and return slightly smaller values.